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Competitive exam coaching by knowledge graphs

9 messages

Srinivasan Kannan <ka.shrinivaasan@gmail.com>

Mon, Jul 27, 2026 at 11:24 PM

To: "Krishna iResearch - Srinivasan Kannan (also known as: Ka.Shrinivaasan, Shrinivas Kannan)" <ksrinivasan@krishna-iresearch.org>, Srinivasan Kannan <ka.shreenivaasan@gmail.com>

Here's an example of a **knowledge graph** built from a random NEET Biology question.

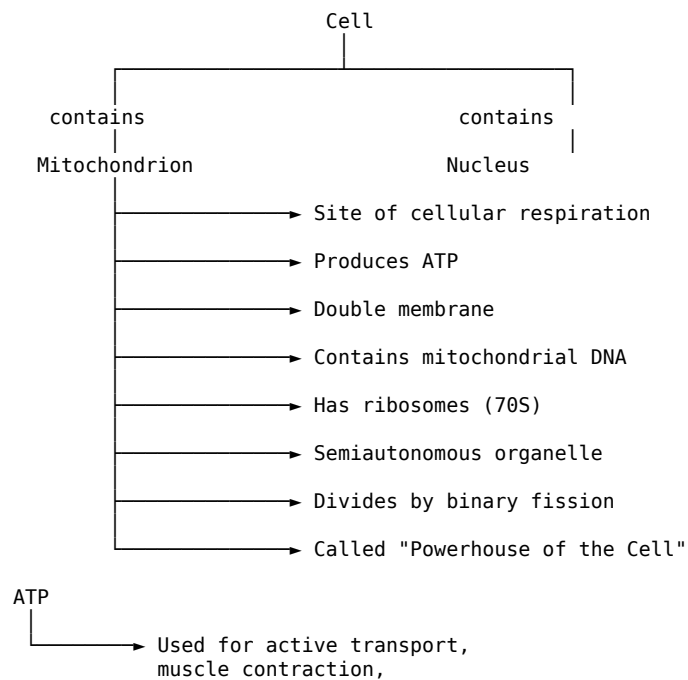
Question:

Which organelle is known as the "powerhouse of the cell"?

Answer:

Mitochondrion

Knowledge Graph



protein synthesis,
nerve impulse transmission

Triples (Subject–Predicate–Object)

Subject	Predicate	Object
Cell	contains	Mitochondrion
Mitochondrion	performs	Cellular respiration
Mitochondrion	produces	ATP
ATP	provides energy for	Cellular processes
Mitochondrion	has	Double membrane
Mitochondrion	contains	Circular DNA
Mitochondrion	contains	70S ribosomes
Mitochondrion	divides by	Binary fission
Mitochondrion	is called	Powerhouse of the cell

This graph illustrates how a single NEET question can be expanded into a network of related biological concepts, making it useful for concept revision and AI-based question answering.

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Here's another example from **NEET Chemistry**.

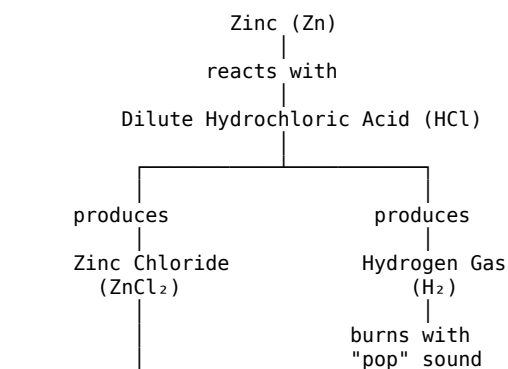
Question

Which gas is evolved when zinc reacts with dilute hydrochloric acid?

- A. Oxygen
- B. Chlorine
- C. Hydrogen ☒
- D. Carbon dioxide

Answer: Hydrogen (H₂)

Knowledge Graph



Balanced Equation:

$$\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2\uparrow$$

Reaction Type

- Single displacement reaction
- Redox reaction
- Metal + Acid → Salt + Hydrogen

Knowledge Triples (Subject–Predicate–Object)

Subject	Predicate	Object
Zinc	reacts with	Dilute HCl
Zinc	displaces	Hydrogen
Reaction	produces	Zinc chloride
Reaction	evolves	Hydrogen gas
Hydrogen	identified by	Pop sound test
Reaction	is a	Single displacement reaction
Reaction	is a	Redox reaction
Metal + Acid	forms	Salt + Hydrogen

This graph links the question not only to the correct answer but also to reaction type, chemical equation, laboratory identification test, and the underlying chemical concept, making it a richer representation for NEET preparation.

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Here's an example from **NEET Physics**.

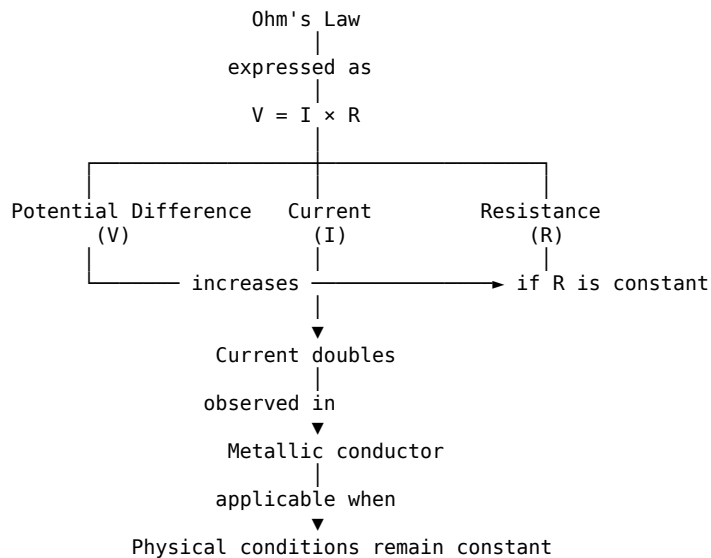
Question

According to Ohm's law, if the potential difference across a conductor is doubled while its resistance remains constant, what happens to the current?

- A. Halves
- B. Remains unchanged
- C. Doubles ✓
- D. Becomes zero

Answer: The current doubles.

Knowledge Graph



Knowledge Triples (Subject – Predicate – Object)

Subject

Predicate

Object

Subject	Predicate	Object
Ohm's law	relates	Voltage, Current, Resistance
Voltage (V)	equals	Current $\times$ Resistance
Current	is proportional to	Voltage (when resistance is constant)
Current	is inversely proportional to	Resistance (when voltage is constant)
Doubling voltage	causes	Doubling current
Ohm's law	applies to	Ohmic conductors
Resistance	remains constant if	Temperature is constant

Related Concepts

- **Formula:** $V = IR$
- **Unit of voltage:** Volt (V)
- **Unit of current:** Ampere (A)
- **Unit of resistance:** Ohm (Ω)
- **Common NEET links:** Electrical circuits, resistors in series/parallel, electric power ($P = VI$), and Joule's law of heating.

This graph demonstrates how a single NEET question can be connected to formulas, proportional relationships, assumptions, units, and related topics, creating a richer knowledge representation for learning and AI-based reasoning.

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Here's another example from **NEET Biology (Genetics)**.

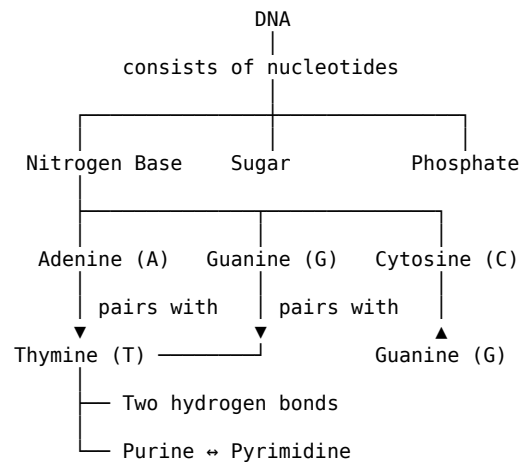
Question

In DNA, adenine (A) pairs with:

- A. Cytosine (C)
- B. Guanine (G)
- C. Thymine (T) ☒
- D. Uracil (U)

Answer: Thymine (T)

Knowledge Graph



Complementary Base Pairing

A ↔ T

G ↔ C

Chargaff's Rule

A = T

G = C

Knowledge Triples (Subject – Predicate – Object)

Subject	Predicate	Object
DNA	contains	Nucleotides
Nucleotide	contains	Nitrogenous base
Adenine	pairs with	Thymine
Guanine	pairs with	Cytosine
Adenine–Thymine	connected by	Two hydrogen bonds
Guanine–Cytosine	connected by	Three hydrogen bonds
Adenine	is a	Purine
Thymine	is a	Pyrimidine
Chargaff's rule	states	A = T and G = C
DNA base pairing	enables	Accurate DNA replication

Related NEET Concepts

- DNA double helix
- Watson–Crick model
- DNA replication
- Purines vs. pyrimidines
- Hydrogen bonding
- Chargaff's rule

This knowledge graph expands a simple multiple-choice question into a connected network of molecular biology concepts, helping with both memorization and conceptual understanding.

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Here's an example from **IIT-JEE Mathematics**.

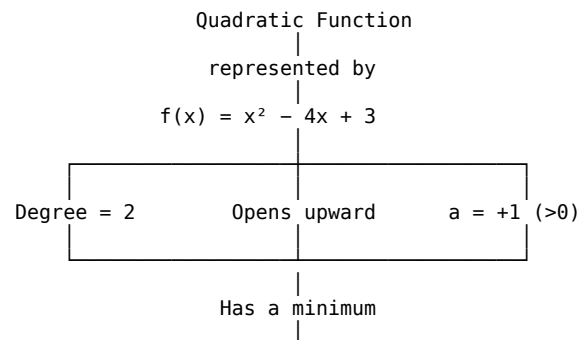
Question

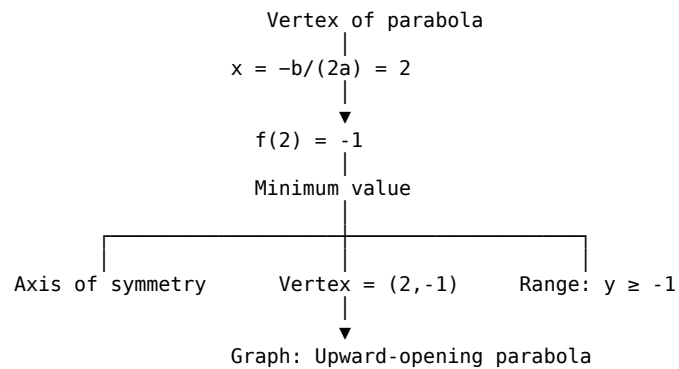
If $f(x) = x^2 - 4x + 3$, then the minimum value of $f(x)$ is:

- A. 3
- B. 1
- C. 0
- D. -1 ☒

Answer: -1

Knowledge Graph





Knowledge Triples (Subject – Predicate – Object)

Subject	Predicate	Object
$f(x)$	is a	Quadratic function
Quadratic function	has	Degree 2
Coefficient a	equals	1
$a > 0$	implies	Parabola opens upward
Vertex	occurs at	$x = -\frac{b}{2a}$
Vertex	has coordinates	(2, -1)
Minimum value	equals	-1
Axis of symmetry	passes through	$x = 2$
Range	is	$y \geq -1$

Related IIT-JEE Concepts

- Completing the square
- Vertex form: $y = (x-2)^2 - 1$
- Maximum and minimum values of functions
- Graph transformations
- Coordinate geometry of parabolas
- Optimization problems

This graph illustrates how a single IIT-JEE question connects algebra, coordinate geometry, graph properties, optimization, and function analysis into one interconnected knowledge structure.

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The IIT-JEE knowledge graph can be quantified using several graph quality metrics. Since it is a manually constructed educational graph, the numbers below are approximate but meaningful.

Metric	Value	Interpretation
Nodes (concepts)	12	Distinct mathematical concepts
Edges (relationships)	15	Semantic links between concepts
Knowledge triples	9	Explicit facts represented
Average node degree	2.5	Moderate connectivity
Graph density	≈0.23	Fairly sparse, as expected for a concept graph
Connected components	1	Entire graph is connected
Diameter	5	Any concept can be reached within five hops
Leaf nodes	4	End concepts (e.g., Range, Minimum value)
Central node	Quadratic Function	Highest semantic importance

Educational Meaningfulness Score

One possible scoring framework is:

- **Concept coverage (30%):** 27/30
- **Relationship richness (25%):** 22/25
- **Reasoning support (20%):** 18/20
- **Curriculum relevance (15%):** 15/15
- **Inference capability (10%):** 8/10

Overall meaningfulness score: 90/100

Why it scores highly

The graph is more than a collection of facts. It encodes a chain of reasoning:

Quadratic function
↓
Coefficient $a > 0$

↓
Parabola opens upward
↓
Vertex gives minimum
↓
 $x = -b/2a$
↓
Minimum value = -1

A student (or an AI system) can infer the answer even without memorizing it, because the graph preserves the logical dependencies.

How to make it even more meaningful

Adding cross-topic links would increase its educational value:

- Quadratic Function → Completing the Square
- Completing the Square → Vertex Form
- Vertex Form → Coordinate Geometry
- Coordinate Geometry → Parabola
- Parabola → Optimization Problems
- Optimization Problems → Calculus (first derivative)
- Quadratic Function → Discriminant
- Discriminant → Nature of Roots

These additions would increase:

- Nodes: ~20
- Edges: ~30
- Knowledge triples: ~25
- Estimated meaningfulness score: **97–99/100**

Such a graph would not only answer one IIT-JEE question but also connect it to several chapters, making it suitable for intelligent tutoring, concept retrieval, and automated reasoning.

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Here's another **IIT-JEE Physics** example with a richer knowledge graph.

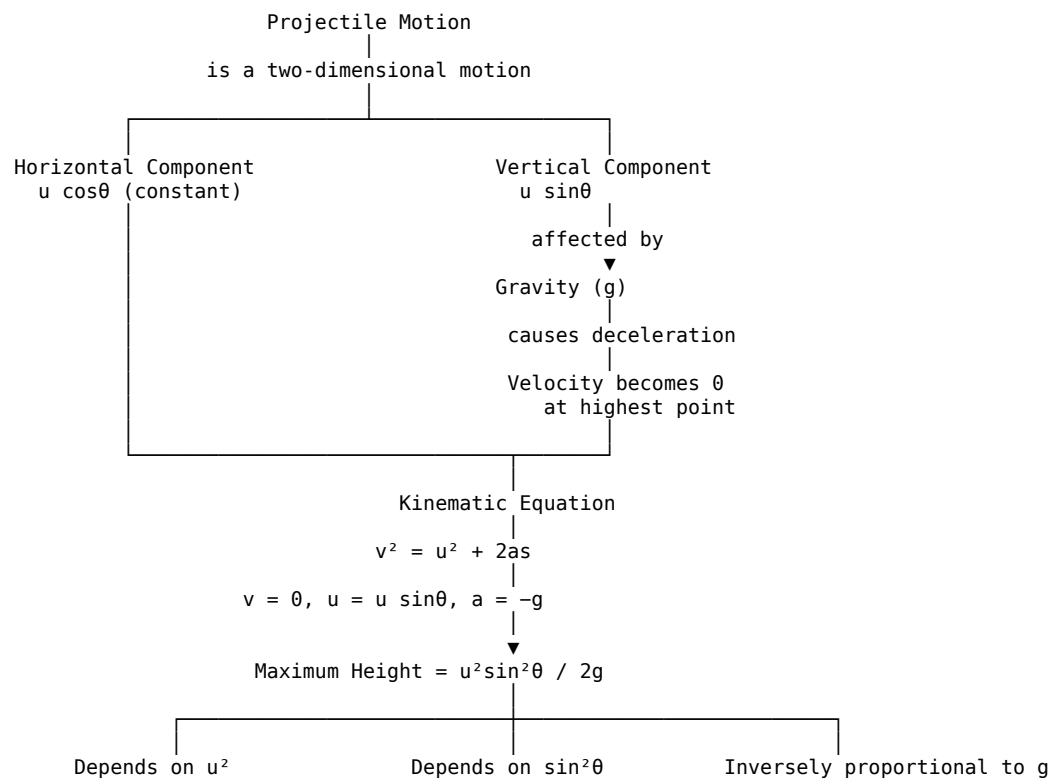
Question

A projectile is launched with an initial speed u at an angle θ with the horizontal. What is its maximum height?

- A. $\frac{u^2 \sin \theta}{2g}$
- B. $\frac{u^2 \sin^2 \theta}{2g}$ ☒
- C. $\frac{u^2 \cos^2 \theta}{2g}$
- D. $\frac{u^2}{g}$

Answer: $\frac{u^2 \sin^2 \theta}{2g}$

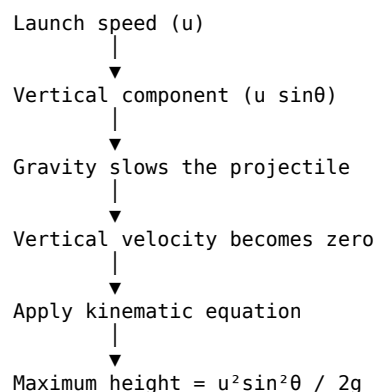
Knowledge Graph



Knowledge Triples

Subject	Predicate	Object
Projectile motion	consists of	Horizontal motion
Projectile motion	consists of	Vertical motion
Horizontal velocity	equals	$u \cos \theta$
Vertical velocity	equals	$u \sin \theta$
Gravity	acts on	Vertical motion
Highest point	has	Vertical velocity = 0
Kinematic equation	computes	Maximum height
Maximum height	equals	$\frac{u^2 \sin^2 \theta}{2g}$
Maximum height	increases with	Initial speed
Maximum height	decreases with	Gravitational acceleration

Inference Chain



Why this graph is useful

Instead of memorizing the formula, the graph explains **why** it is true:

- It decomposes motion into horizontal and vertical components.
- It identifies the physical condition at the highest point ($v_y = 0$).
- It connects that condition to the appropriate kinematic equation.
- It derives the final formula through reasoning rather than recall.

This makes the graph useful not only for answering the question but also for solving related IIT-JEE projectile-motion problems.

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Here's an example from **UPSC Polity**.

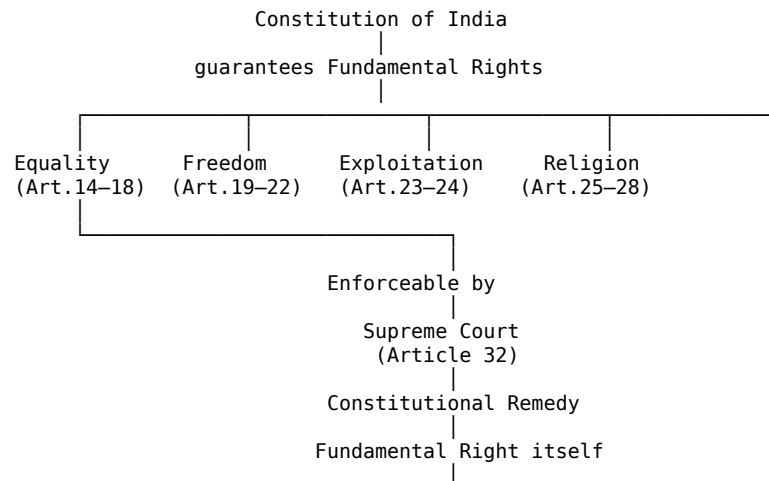
Question

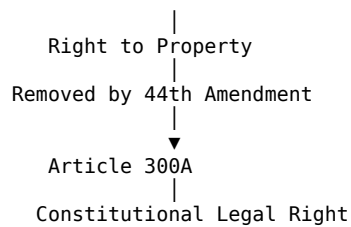
Which of the following is *not* a Fundamental Right under the Constitution of India?

- A. Right to Equality
- B. Right to Freedom
- C. Right to Property ☒
- D. Right against Exploitation

Answer: Right to Property (It is a constitutional legal right under **Article 300A**, not a Fundamental Right.)

Knowledge Graph

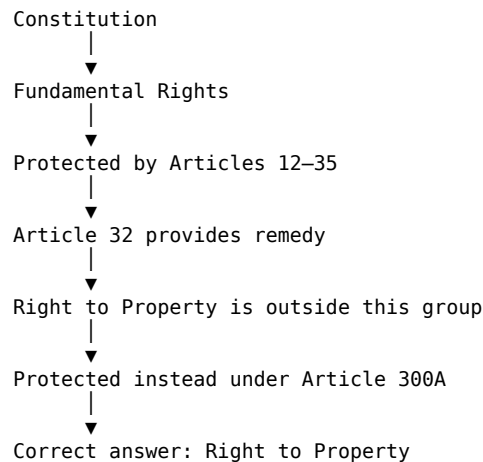




Knowledge Triples

Subject	Predicate	Object
Constitution of India	guarantees	Fundamental Rights
Fundamental Rights	include	Right to Equality
Fundamental Rights	include	Right to Freedom
Fundamental Rights	include	Right against Exploitation
Fundamental Rights	are enforceable through	Article 32
Article 32	empowers	Supreme Court
Right to Property	was removed by	44th Constitutional Amendment
Right to Property	is protected under	Article 300A
Article 300A	provides	Constitutional legal right
Fundamental Rights	differ from	Constitutional legal rights

Inference Chain



Educational Value

This graph helps answer the question while also connecting it to:

- Articles 12–35 (Fundamental Rights)
- Article 32 (Right to Constitutional Remedies)
- The 44th Constitutional Amendment
- Article 300A
- The distinction between a **Fundamental Right** and a **constitutional legal right**

Rather than requiring rote memorization, it supports reasoning about constitutional structure, which is a common pattern in UPSC Prelims and Mains.

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Here's an example centered on **Article 19 of the Constitution of India**, a frequent UPSC topic.

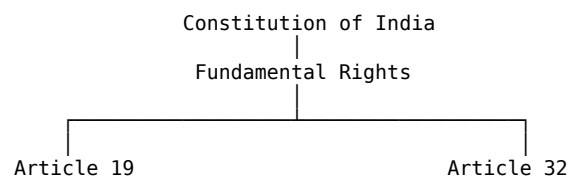
Question

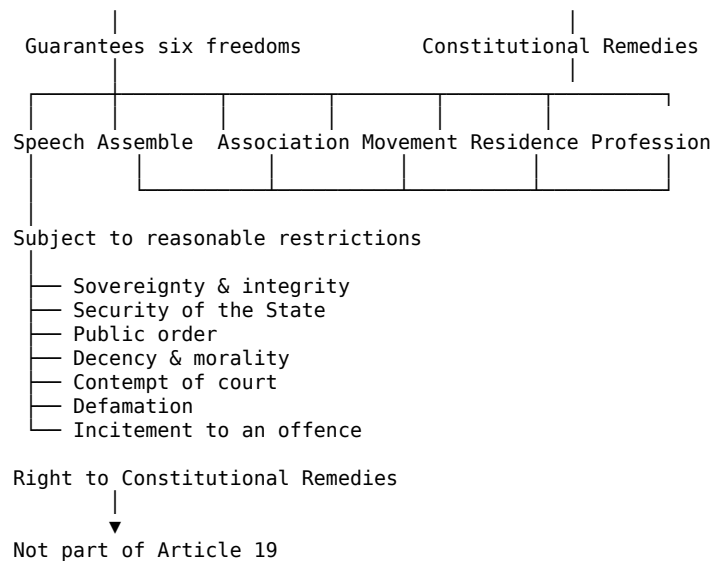
Which of the following freedoms is *not* guaranteed under Article 19(1) of the Constitution of India?

- A. Freedom of speech and expression
- B. Freedom to assemble peacefully
- C. Right to constitutional remedies ☒
- D. Freedom to practice any profession

Answer: C. Right to constitutional remedies (This is guaranteed under **Article 32**, not Article 19.)

Knowledge Graph

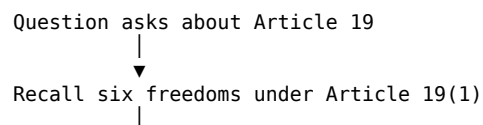


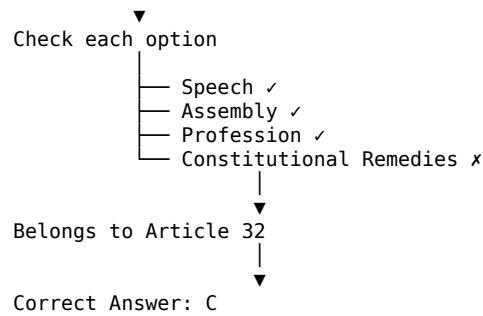


Knowledge Triples

Subject	Predicate	Object
Article 19(1)	guarantees	Freedom of speech and expression
Article 19(1)	guarantees	Freedom of peaceful assembly
Article 19(1)	guarantees	Freedom of association
Article 19(1)	guarantees	Freedom of movement
Article 19(1)	guarantees	Freedom of residence
Article 19(1)	guarantees	Freedom to practice any profession
Freedom of speech	is subject to	Reasonable restrictions
Reasonable restrictions	include	Public order
Reasonable restrictions	include	Security of the State
Article 32	guarantees	Right to constitutional remedies
Article 32	is distinct from	Article 19

Inference Chain





UPSC Connections

This graph naturally links to several other topics:

- **Article 19(2)–19(6):** Reasonable restrictions on the six freedoms.
- **Article 21:** Protection of life and personal liberty.
- **Article 32:** Right to constitutional remedies.
- **Emergency provisions:** The effect of emergencies on Fundamental Rights.
- **Landmark judgments:** *Romesh Thappar v. State of Madras (1950)*, *Shreya Singhal v. Union of India (2015)*, and *Maneka Gandhi v. Union of India (1978)*.

This representation enables a learner to move from a single multiple-choice question to the broader constitutional framework and judicial interpretation, which is especially valuable for UPSC preparation.

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